

ERW: Ex-Ante Modelling in Sub-Saharan Africa

**An end-to-end technical report on the modelling approach:
workflow, equations, assumptions, and results**

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About this report

This report documents, end to end, the ex-ante model that maps where Enhanced Rock Weathering (ERW) with crushed basalt is likely to be worthwhile on Sub-Saharan African (SSA) cropland. It explains the *workflow* (which script computes what, in what order), the *equations* (every mechanism and constant, drawn directly from the code), the *modelling choices* (why the model is shaped the way it is, and where it uses defensible proxies), and the *results* produced under the current defaults.

What this model is. It is a spatially explicit *screening* model. It ranks and compares locations, quantifies the two returns ERW can generate — durable carbon dioxide removal (a public good) and an agronomic liming co-benefit (a private good) — and stress-tests those returns against price, yield, and logistics assumptions. It is deliberately transparent and fast: every term is an interpretable multiplier that can be traced to a published anchor.

What this model is not. It is not a monitoring, reporting and verification (MRV) or carbon-crediting instrument. The absolute per-hectare tonnages it reports are order-of-magnitude estimates, not bankable credits; credit-grade numbers require mechanistic reactive-transport modelling and field measurement ([chapter 11](#)). Read the tonnages as a way to compare pixels, not to certify them.

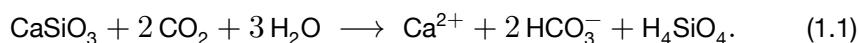
Relationship to the other documents. Two companion documents go deeper on specific pieces: the *Carbon Dioxide Removal — Technical Derivation* ([erw-cdr-calculations.pdf](#)) sources every step of the CDR stoichiometry and net-export accounting; the *parameters and assumptions* catalogue ([erw-parameters.md](#)) is the master list of constants with citations. This report is the integrating narrative that ties the whole pipeline together.

Headline numbers are quoted from the model run dated 2026-05-26 unless stated otherwise. Where a worked example uses earlier defaults, this is flagged; the equations are unchanged, only the input constants differ.

1. Introduction and scope

1.1 What Enhanced Rock Weathering does

Silicate rocks weather naturally: rain-derived carbonic acid attacks the minerals, the acid is consumed, and the carbon leaves as dissolved bicarbonate. Over geological time this is one of Earth's principal thermostats. Enhanced Rock Weathering simply accelerates it — grind a fast-weathering silicate such as basalt to a powder, spread it on a field, and the vastly increased mineral surface area lets the reaction run in years rather than millennia ([Beerling et al., 2020](#); [Hartmann et al., 2013](#)). For a divalent cation such as calcium, the net reaction consumes two moles of CO₂ per mole of cation released:



The captured carbon is durable for 10³–10⁵ years once it reaches the ocean as bicarbonate ([Kanzaki, Planavsky, & Reinhard, 2023](#); [Renforth, 2019](#)).

ERW is attractive on *cropland* for a second reason. The alkalinity that basalt releases is chemically the same medicine that agricultural lime delivers: it raises pH and neutralises the exchangeable acidity and aluminium toxicity that suppress yields on the acidic soils common across the tropics. So a single application can pay twice — a durable carbon credit (the *public* return) and a yield uplift the farmer keeps (the *private* return). Much of SSA has the acidic, weathered soils, the warm-wet climate that speeds weathering, and the smallholder yield gap that makes this pairing compelling ([Baek et al., 2023](#); [Beerling et al., 2020](#)).

1.2 The question the model answers

Given all of the above, *where* on SSA cropland is ERW worth doing, for whom, and how sensitive is that answer to what we assume? The model answers this pixel by pixel and crop by crop, holding the biogeochemistry, agronomy, and economic geography in one accounting identity ([chapter 2](#)).

1.3 Scope and units

At a glance. ~9 km working grid (SoilGrids-Africa aggregated $10\times$), 47 SSA countries, 23 SPAM crops, all monetary values in USD, a 5-year project horizon, and a per-pixel / farmer's-eye accounting perspective.

Table 1.1: Spatial, temporal, and accounting scope of the model.

Dimension	Setting
Region	Sub-Saharan Africa, 47 countries (GADM v4); islands excluded
Working grid	SoilGrids-Africa 30'' aggregated $10\times \approx 0.083^\circ$ (~9 km)
Land mask	QED cropland, pixels with $> 10\%$ cropland fraction
Crops	23 SPAM v2 crops (cereals, legumes, commodities, roots & tubers)
Horizon	5 years (industry accreditation window)
Currency	USD throughout
Perspective	Per pixel, farmer's-eye; no aggregator-level finance
Feedstock	Basalt (mafic silicate); ultramafics excluded (Ni/Cr risk)

The remainder of the report follows the model's own logic: the scientific framework ([chapter 2](#)), the pipeline and data ([chapter 3](#)), then each computational block — feedstock and rate ([chapter 4](#)), CDR ([chapter 5](#)), crop response ([chapter 6](#)), cost and logistics ([chapter 7](#)), profitability ([chapter 8](#)), and supply and sensitivity ([chapter 9](#)) — and closes with results ([chapter 10](#)), limitations and roadmap ([chapter 11](#)), and reproducibility ([chapter 12](#)).

2. Scientific framework

The model decomposes the value of a basalt application into three independent “streams” that are computed separately and then reconciled in a single per-pixel gross-margin identity (Figure 2.1).

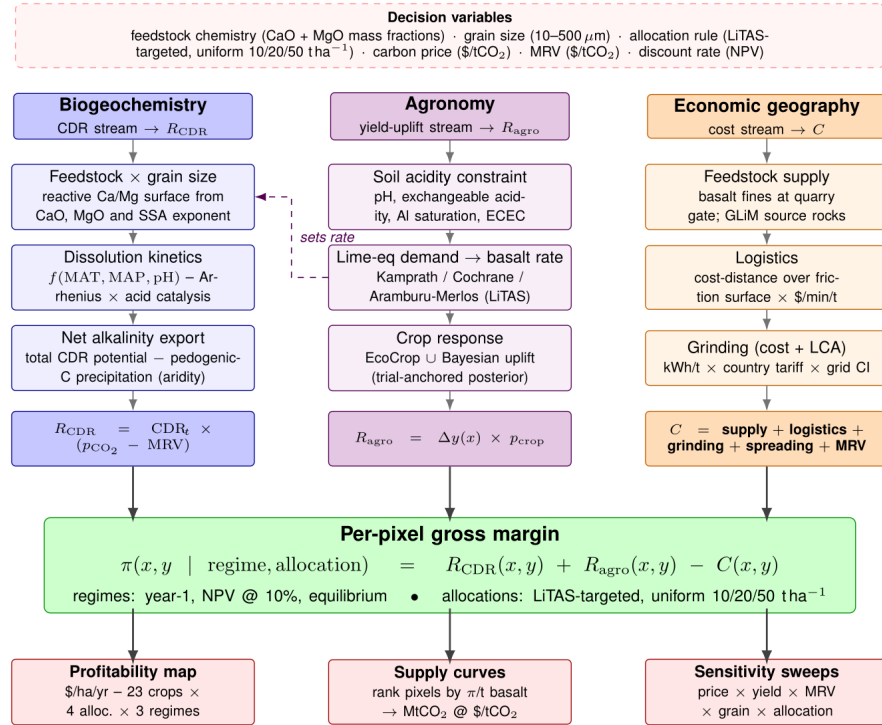


Figure 2.1: The three-stream conceptual framework. A biogeochemical stream turns basalt into durable CDR (R_{CDR}); an agronomic stream turns the same alkalinity into a yield uplift (R_{agro}); an economic-geography stream builds up the delivered-and-applied cost (C). The dashed cross-arrow marks the coupling: the agronomic lime demand *sets the application rate* that both other streams then scale.

2.1 The three streams

Biogeochemistry $\rightarrow R_{\text{CDR}}$. From feedstock chemistry and grain size, through dissolution kinetics that respond to temperature, moisture and pH, to the net alkalinity actually exported (total capture minus the fraction that re-precipitates locally as pedogenic carbonate). Detailed in [chapter 4](#) and [chapter 5](#).

Agronomy $\rightarrow R_{\text{agro}}$. From the soil acidity constraint (pH, exchangeable acidity, aluminium saturation, ECEC) to a lime-equivalent demand, which both sets the basalt rate and — via a crop suitability response — implies a yield uplift. Detailed in [chapter 6](#).

Economic geography $\rightarrow C$. From feedstock supply at the quarry gate, through cost-distance logistics over a friction surface, grinding energy priced at the local electricity tariff, spreading, and MRV. Detailed in [chapter 7](#).

2.2 The master identity

For a pixel at location x growing crop y , under a chosen accounting regime and allocation rule, the per-pixel gross margin is

$$\pi(x, y \mid \text{regime, allocation}) = R_{\text{CDR}}(x, y) + R_{\text{agro}}(x, y) - C(x, y). \quad (2.1)$$

Everything downstream — profitability maps, supply curves, sensitivity sweeps — is a transformation of [Equation 2.1](#). The *decision variables* (the dials a project designer can turn) are feedstock chemistry, grain size, allocation rule (targeted lime-requirement dose vs. uniform 10/20/50 t ha⁻¹), carbon price, MRV cost, and discount rate.

Why the streams are coupled, not independent. The agronomic lime-requirement calculation produces the *rate* (t ha⁻¹) of basalt. That single rate then multiplies both the CDR per hectare and the cost per hectare. Applying more basalt buys more carbon and more yield relief, but also more cost and more lifecycle emissions — which is why the “right” rate is a genuine optimisation, not a maximisation ([chapter 9](#)).

3. Pipeline architecture and data

3.1 Nine blocks, one dependency graph

The pipeline is implemented as a chain of R scripts (prefix `erw-`) organised into nine functional blocks, B1–B9 (Figure 3.1). Each script reads rasters written by its predecessors and writes rasters for its successors; there is no hidden state.

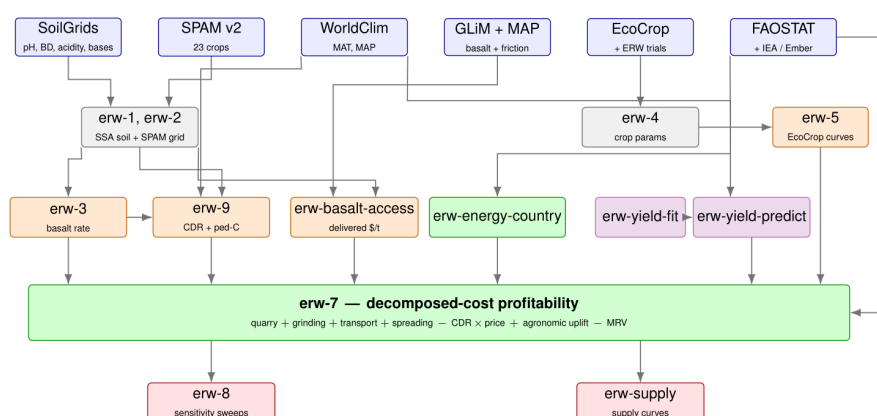


Figure 3.1: Pipeline architecture. Blue = raw data sources; grey = the common working grid; orange = geospatial surfaces; green = energy; purple = the (optional) Bayesian yield arm; the wide green bar = the profitability engine (`erw-7`); red = deliverables. Arrows are data dependencies.

Table 3.1: The nine blocks and the scripts that implement them.

Block	Script(s)	What it produces
B1 Grid	<code>erw-1</code> , <code>erw-2</code>	SSA + cropland mask; depth-weighted soil stack; SPAM area/production/yield for 23 crops

Block	Script(s)	What it produces
B2 Rate	erw-3	Lime-equivalent demand → basalt rate; feedstock chemistry; uniform-rate rasters
B6 Crop	erw-4, erw-5	Per-crop EcoCrop pH and acidity tolerance tables and relative-yield rasters
B3 CDR	erw-9	Climate/pH-scaled, pedogenic-deducted, net-export CDR per hectare
B4 Logistics	erw-basalt-access (+ build-basalt-mask, fetch-friction-surface)	Basalt sources; travel time; delivered price
B5 Energy	erw-energy-country	Country electricity tariff and grid carbon intensity
(B6') Bayesian	erw-yield-fit, erw-yield-predict	Trial-anchored per-crop uplift rasters (optional; not run at v1)
B7 Profit	erw-7	Decomposed-cost gross margin: 4 allocations × 3 regimes × 23 crops = 276 rasters
B8 Supply	erw-supply	Pixel ranking by GM per tonne basalt; supply curves
B9 Sensitivity	erw-8	Five parameter sweeps

3.2 Data sources

Table 3.2: External datasets and their roles.

Dataset	Provides	Source
SoilGrids-Africa	pH, bulk density, exch. acidity, exch. bases (K/Ca/Mg/Na)	Poggio et al. (2021)
SPAM v2	Harvested area, production, yield, 23 crops	Yu et al. (2020)
WorldClim 2.1	Mean annual temperature, precipitation	Fick and Hijmans (2017)
GLiM v1	Basalt-source lithology (basic volcanic, pyroclastic, gabbro)	Hartmann and Moosdorf (2012)
MAP friction surface	Motorised travel-time friction	Weiss et al. (2020)

Dataset	Provides	Source
EcoCrop / Recocrop	Crop pH and acidity tolerance envelopes	Hijmans (2023)
FAOSTAT	Producer prices (2016–2020 SSA median)	FAO (2023)
Ember / IEA	Grid carbon intensity, electricity tariffs	country table
CGIAR-CSI	Global Aridity Index (pedogenic deduction)	Zomer, Xu, and Trabucco (2022)

3.3 Building the working grid (B1)

erw-1 defines SSA (GADM v4, dropping island states), applies the QED cropland filter (keep pixels $> 10\%$ cropland), and assembles the SoilGrids soil-property stack. Soil properties are *depth-weighted* over the top 30 cm,

$$p = \frac{5p_5 + 10p_{15} + 15p_{30}}{30}, \quad (3.1)$$

and two derived layers are formed inline: the effective cation exchange capacity $ECEC = H_{\text{exch}}^+ + \sum (K, Ca, Mg, Na)$ and the acidity saturation $H_{\text{sat}}^+ = 100 H_{\text{exch}}^+ / ECEC$ (%). erw-2 pulls SPAM v2 for all 23 crops and resamples everything onto the common $10\times$ -aggregated grid, so every later raster is co-registered.

4. Feedstock chemistry and the basalt rate

This block (erw-3) answers two coupled questions: *how much* basalt to apply (the rate), and *how much carbon* a tonne of it could remove (the per-tonne ceiling, before spatial de-rating in [chapter 5](#)).

4.1 Feedstock composition

Basalt is represented by a small set of oxide mass fractions ([Table 4.1](#)); only CaO and MgO are consumed in the current CDR stoichiometry (K, Na, P are carried for future nutrient accounting; Ni and Cr are carried as a contamination-risk surrogate).

Table 4.1: Default basalt chemistry (mass fractions unless noted).

Component	Default	Role
CaO	0.10	Cation source (consumed)
MgO	0.07	Cation source (consumed)
K ₂ O	0.015	Nutrient (not yet consumed)
Na ₂ O	0.025	Carried, not consumed
P ₂ O ₅	0.003	Nutrient (not yet consumed)
Ni	100 ppm	Contamination-risk surrogate
Cr	200 ppm	Contamination-risk surrogate
Grain size d	50 μm	Decision variable
Reactive fraction ϕ_0	0.20	Fraction of stoichiometric limit realised

4.2 Stoichiometric ceiling

Using standard oxide-to-product mass ratios (1 g CaO $\rightarrow$ 1.783 g CaCO₃-eq and 1.570 g CO₂; 1 g MgO $\rightarrow$ 2.481 g CaCO₃-eq and 2.185 g CO₂), the calcium-carbonate equivalent and the stoichiometric CDR ceiling per tonne of

basalt are

$$\text{CCE} = 0.10 \times 1.783 + 0.07 \times 2.481 \approx 0.352 \quad (352 \text{ kg CaCO}_3/\text{t}), \quad (4.1)$$

$$\text{CDR}_{\text{max}} = (0.10 \times 1.570 + 0.07 \times 2.185) \times 1000 \approx \mathbf{310} \text{ kg CO}_2/\text{t basalt}. \quad (4.2)$$

Equation 4.2 is a hard upper bound — the carbon captured if every calcium and magnesium ion weathered and every bicarbonate reached the ocean. Reality is a fraction of it, and quantifying that fraction is the whole task of chapter 5.

4.3 Grain size: a reactivity–energy trade-off

Grinding finer exposes more surface area (more reactivity) but costs more energy (more cost and lifecycle emissions). Both sides are power laws anchored to the literature. The reactivity (surface-area) factor, relative to a 100 μm reference, uses exponent $\beta = 0.5$:

$$f_{\text{grain}} = \left(\frac{100}{d}\right)^{0.5}, \quad f_{\text{grain}}(50 \mu\text{m}) = \sqrt{2} \approx 1.414. \quad (4.3)$$

Grinding energy follows the Streffer curve (0.07 GJ/t at 50 μm , exponent $\alpha = 1.2$; Streffer, Amann, Bauer, Kriegler, & Hartmann, 2018):

$$E_{\text{grind}}(d) = 0.07 \left(\frac{50}{d}\right)^{1.2} \text{ GJ/t}, \quad \text{kWh/t} = E_{\text{grind}} \times 277.78. \quad (4.4)$$

At the 50 μm default this is 19.4 kWh/t. Grain size is a decision variable precisely because Equation 4.3 and Equation 4.4 pull in opposite directions; chapter 9 sweeps it.

4.4 From lime demand to basalt rate

The *effective neutralising value* folds the reactive fraction and grain factor into the CCE, and its reciprocal converts a lime requirement into a basalt rate:

$$\text{NV}_{\text{eff}} = \text{CCE} \cdot \phi_0 \cdot f_{\text{grain}}, \quad r_{\text{bl}} = \frac{1}{\text{NV}_{\text{eff}}}. \quad (4.5)$$

At the 50 μm default, $\text{NV}_{\text{eff}} \approx 0.10$ so $r_{\text{bl}} \approx 10.0$ t basalt per t CaCO_3 -equivalent. The reference per-tonne CDR that chapter 5 scales spatially is $\text{CDR}_{\text{eff}} = \text{CDR}_{\text{max}} \cdot \phi_0 \cdot f_{\text{grain}} \approx 88 \text{ kg CO}_2/\text{t}$.

The lime requirement L itself comes from the Limer engine, which implements three published tropical-soil methods — Kamprath, Cochrane, and the

Aramburu-Merlos LiTAS (“lime to a target acidity saturation”) model (Aramburu Merlos, Silva, Baudron, & Hijmans, 2023). The targeted basalt rate is

$$R_{\text{targeted}} = L \times r_{\text{bl}} \quad (\text{t basalt ha}^{-1}), \quad (4.6)$$

applied only where acidity saturation exceeds a threshold ($> 10\%$, and, per crop, above the crop’s own tolerance; chapter 6). The counterfactual *uniform* rules apply a flat 10, 20, or 50 t ha^{-1} everywhere there is cropland, matching the convention of continental studies (Baek et al., 2023; Beerling et al., 2020). Figure 4.1 shows the resulting LiTAS rate.

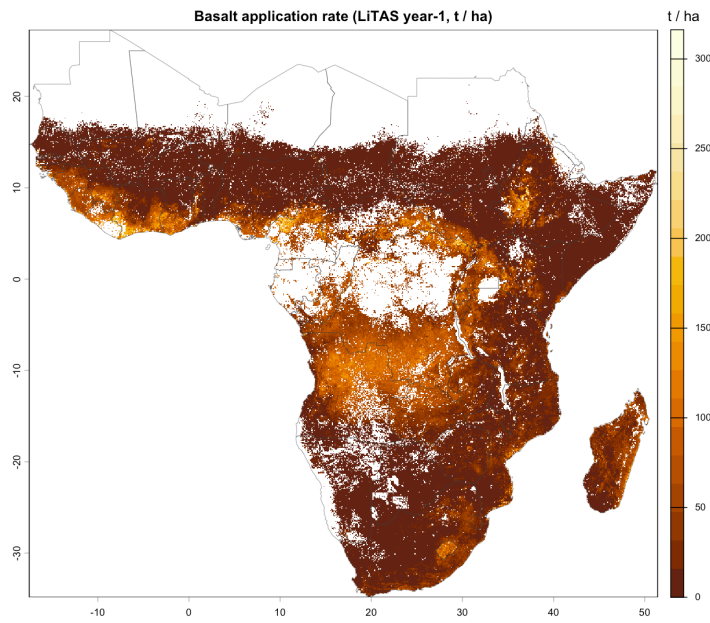


Figure 4.1: Targeted (LiTAS) basalt application rate, t ha^{-1} . High where soils are strongly acidic and lime demand is large; zero on near-neutral soils, which the targeted rule skips.

5. The carbon-dioxide-removal surface

The stoichiometric ceiling of Equation 4.2 is de-rated, pixel by pixel, by three physical realities: not all of the mineral reacts (reactive fraction, scaled by climate), some of the released alkalinity re-precipitates locally as carbonate (pedogenic loss), and some is spent neutralising the soil's own acidity rather than being exported (net-export correction). The cascade is summarised in Figure 5.1.

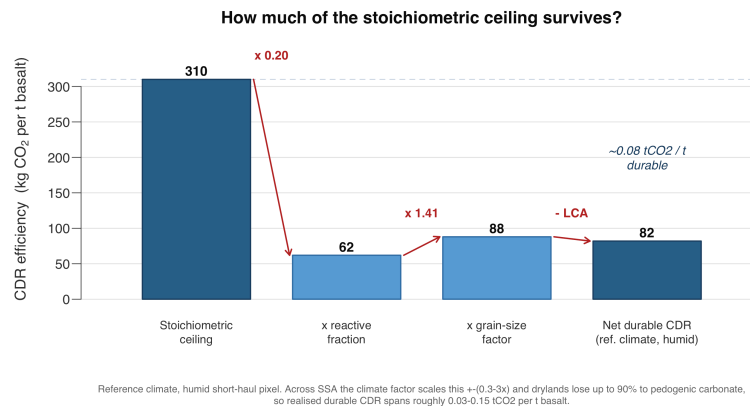


Figure 5.1: How little of the stoichiometric ceiling survives. At a reference climate, humid, short-haul pixel the 310 kg CO₂/t ceiling falls to ~88 kg gross and ~82 kg net of lifecycle emissions — about 0.08 tCO₂ per tonne of basalt. Across SSA the climate factor scales this by 0.3–3 $\times$ and drylands lose up to 90% to pedogenic carbonate.

5.1 The climate factor

Dissolution speeds up with temperature and moisture and depends on pH. The model uses interpretable multipliers, referenced to the US Corn Belt ($T_{\text{ref}} = 11^{\circ}\text{C}$, $P_{\text{ref}} = 1000\text{ mm}$) where the reactive fraction was calibrated (Lewis et

al., 2021):

$$f_{\text{MAT}} = \text{clamp}\left(\frac{\text{MAT}}{11}, 0.3, 3.0\right), \quad f_{\text{MAP}} = \text{clamp}\left(\frac{\text{MAP}}{1000}, 0.3, 3.0\right), \quad (5.1)$$

$$f_{\text{pH}} = \begin{cases} 0.5 & \text{pH} < 4 \\ 0.5 + 0.5(\text{pH} - 4) & 4 \leq \text{pH} < 5 \\ 1.0 - 0.25(\text{pH} - 5) & 5 \leq \text{pH} < 7 \\ 0.5 & \text{pH} \geq 7 \end{cases} \quad (5.2)$$

$$f_{\text{climate}} = f_{\text{MAT}} \cdot f_{\text{MAP}} \cdot f_{\text{pH}}. \quad (5.3)$$

The pH term (Equation 5.2) is triangular, peaking near pH 5 — acid catalysis is strongest on the acidic soils that also have the most agronomic demand (Brantley, Kubicki, & White, 2008; White & Brantley, 2003). The effective reactive fraction is then capped at unity,

$$\phi_{\text{eff}} = \min(1.0, \phi_0 f_{\text{grain}} f_{\text{climate}}), \quad (5.4)$$

and the per-tonne CDR at a pixel, before the net-export step, is $\text{CDR}_{\text{px}} = \text{CDR}_{\text{max}} \cdot \phi_{\text{eff}} \cdot (1 - p_{\text{ped}})$.

5.2 Pedogenic-carbonate loss

In dry soils, some of the released calcium and magnesium re-precipitates as secondary carbonate, re-releasing the captured CO_2 . The lost fraction p_{ped} is binned by aridity (Aridity Index = MAP/PET, from Zomer et al., 2022, with a MAP-only fallback), following the UNEP desertification classes (Table 5.1). The exported fraction is $1 - p_{\text{ped}}$.

Table 5.1: Pedogenic-carbonate loss fraction by aridity band (Renforth, 2019; United Nations Environment Programme, 1992).

Class	Aridity Index (MAP fallback)	p_{ped}
Hyperarid	< 0.05 (< 120 mm)	0.90
Arid	0.05–0.20 (120–300 mm)	0.70
Semi-arid	0.20–0.50 (300–500 mm)	0.40
Dry sub-humid	0.50–0.65 (500–600 mm)	0.15
Humid	≥ 0.65 (≥ 600 mm)	0.00

5.3 Gross yield and the net-export correction

Gross CDR per hectare is the rate times the per-tonne value, $\text{CDR}_{\text{gross}} = R \cdot \text{CDR}_{\text{px}}/1000$ ($\text{tCO}_2 \text{ ha}^{-1}$, cumulative over the horizon). But alkalinity that neutralises the soil's own exchangeable acidity behaves exactly like lime: the carbon it captured is re-released, so it is not *durable* removal. The net-export correction deducts this acidity sink first:

$$\text{CDR}_{\text{net-export}} = \max(0, \text{CDR}_{\text{gross}} - F \cdot A), \quad F = \frac{\text{CDR}_{\text{eff}}}{\text{NV}_{\text{eff}} \cdot 1000} \approx 0.88, \quad (5.5)$$

where A is the acidity sink expressed as CaCO_3 -equivalent and $F \approx 0.88$ is the CO_2 re-released per tonne of that lime-equivalent. This is a first-order *sequential bound* — it assumes the acidity sink is filled before any carbon is exported — so the true durable CDR lies between Equation 5.5 and the gross yield. Net-export is the recommended default for the public/carbon case (Figure 5.2).

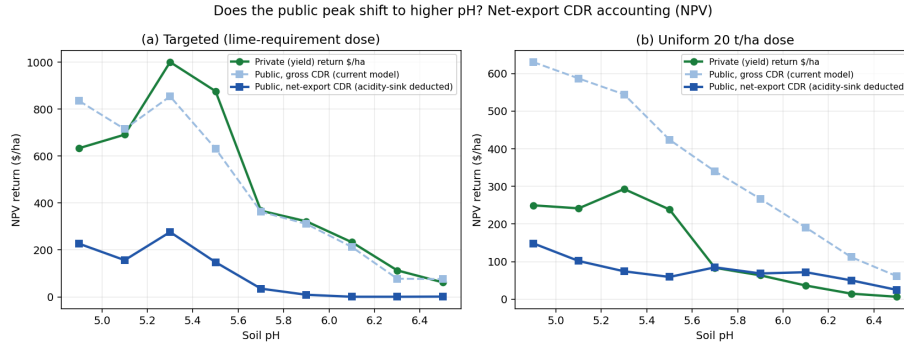
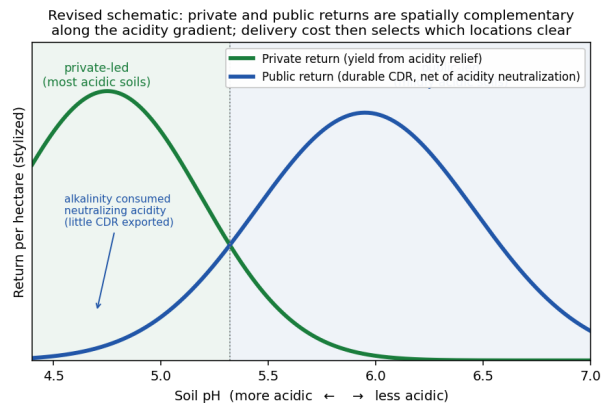


Figure 5.2: Gross vs. net-export CDR along the soil-pH gradient. On the most acidic soils (left), much of the alkalinity is consumed neutralising acidity, so durable (net-export) CDR is far below gross — exactly where the private yield return is largest.

The load-bearing insight. The private (yield) and public (durable CDR) returns peak in *different* places along the acidity gradient. Where soil is very acidic, basalt's alkalinity is spent fixing the soil (great for yield, little exported carbon); where soil is near-neutral, alkalinity is exported (carbon) but there is little yield to gain. The two returns are spatially complementary (Figure 5.3).



Realized returns are additionally gated by the delivery-cost gradient (distance to basalt): within each band, only low-cost, basalt-proximate pixels clear.

Figure 5.3: Private and public returns are spatially complementary along the acidity gradient; delivery cost then selects which locations actually clear.

The spatial pattern of the CDR terms is shown in [Figure 5.4](#). A full, sourced derivation of every constant in this chapter is in the companion *CDR — Technical Derivation*.

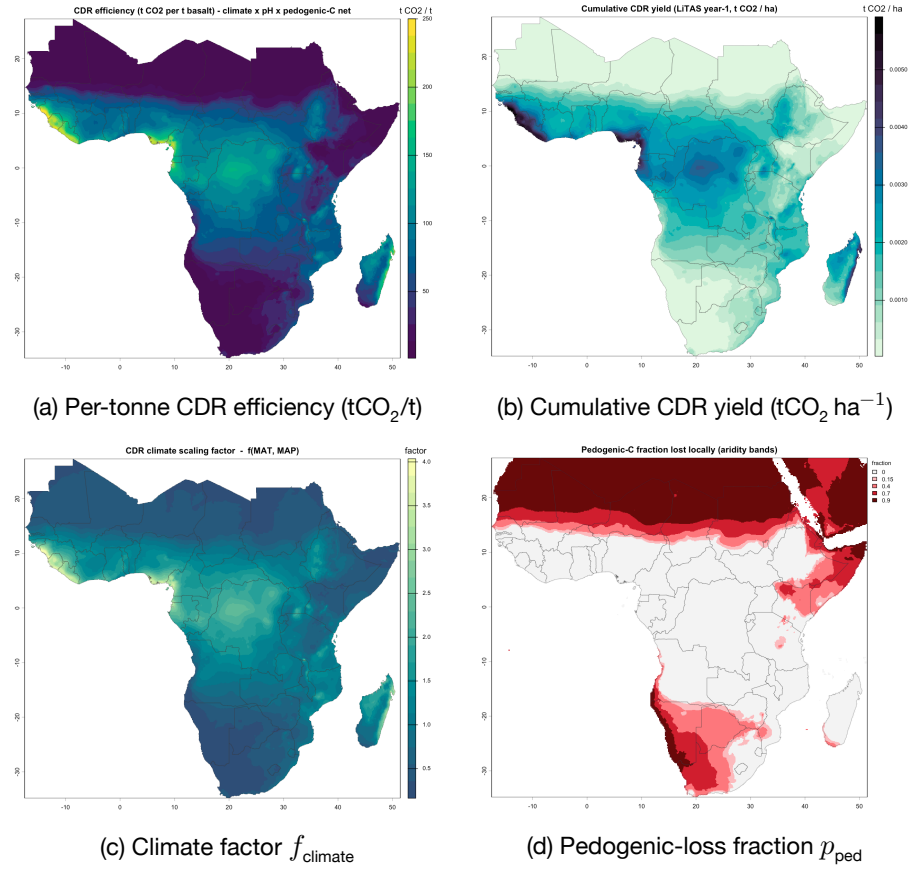


Figure 5.4: The CDR surface and its two spatial modifiers.

6. Crop response: the agro-nomic stream

The agronomic return is the yield uplift a farmer captures when basalt relieves soil acidity, valued at the local producer price. The model offers two ways to estimate that uplift: a deterministic *EcoCrop* suitability model (the default) and an optional trial-anchored *Bayesian* model.

6.1 EcoCrop, conceptually

EcoCrop scores how suitable an environmental condition is for a crop on a 0–1 scale, using a piecewise-linear *tolerance envelope* — a trapezoid with four breakpoints: an absolute minimum below which suitability is zero, an optimal band where it is one, and an absolute maximum above which it is zero again (Figure 6.1a). It is a decades-old agronomic standard, implemented in R as Recocrop (Hijmans, 2023).

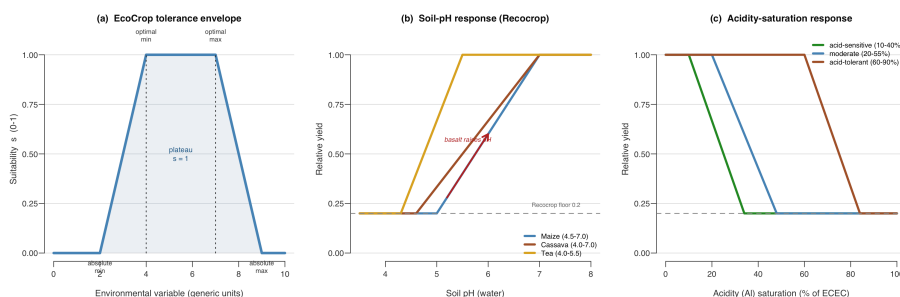


Figure 6.1: The EcoCrop suitability model. (a) The generic trapezoidal tolerance envelope with its four breakpoints. (b) The soil-pH response as parameterised in this pipeline — relative yield rises from a floor to a plateau across each crop’s $[\text{min_ph}, \text{max_ph}]$ band; the red arrow shows how basalt, by raising pH, moves an acidic soil up its curve. (c) The acidity-saturation response — a plateau to the crop’s tolerance, then a linear decline. Both are clamped to a floor of 0.2.

The key modelling insight is that the yield response to soil acidity is the same whether the amendment is lime or basalt — so a pH/acidity suitability model

is a legitimate proxy for basalt's agronomic uplift. In this pipeline the envelope is parameterised two ways per crop:

- **Soil-pH response** (Figure 6.1b): relative yield is a floor below the crop's `min_ph`, rises linearly to 1 at `max_ph`, and plateaus above. Acid-loving crops have low upper bounds (tea 5.5, cocoa 6.0, coffee 6.5); cereals and legumes 7.0–8.0.
- **Acidity-saturation response** (Figure 6.1c): relative yield is 1 up to the crop's aluminium-saturation tolerance `ac_sat` (% of ECEC), then declines linearly to 0 at `max_ac_sat`.

Every raster is clamped to $[0.2, 1]$: no pixel is assigned less than 20% of attainable yield. `erw-5` evaluates these envelopes on the soil stack via `Re-cocrop::ecocrop` to produce 46 rasters ($23 \text{ crops} \times 2 \text{ responses}$).

A modelling choice that unlocked most of SSA. An earlier version used a flat upper pH bound of 5.5 for every crop, which zeroed the agronomic credit across the pH 5.5–7.0 belt that covers most SSA savanna and Sahel. Replacing it with per-crop EcoCrop optima (5.5–8.0) restored a non-zero yield response on that belt — one of the single largest drivers of the refreshed results (chapter 10).

6.2 From suitability to revenue

Inside the profitability engine (`erw-7`) the EcoCrop relative-yield raster loss $\in [0.2, 1]$ is converted to a recoverable tonnage response by dividing the actual yield by the suitability and taking the shortfall,

$$\Delta Y = \frac{Y_0}{\text{loss}} - Y_0, \quad R_{\text{agro}} = \Delta Y \times p_c, \quad (6.1)$$

with p_c the FAOSTAT producer price. The uplift is largest where the soil is acidic *and* the crop is acid-sensitive; it is near zero on already-neutral soils and for acid-tolerant crops (cassava, sorghum, millet), whose response is effectively saturated — which is why those crops must clear the full basalt cost on carbon revenue alone.

6.3 The Bayesian arm (optional, not run at v1)

Where a trial-anchored estimate is preferred, `erw-yield-fit` fits a Bayesian multilevel meta-regression of the log yield-ratio (basalt vs. control) on ERW co-variates, with per-observation known measurement error and crossed crop/site

random intercepts ([Bürkner, 2017](#)):

$$\log\left(\frac{Y_{\text{treat}}}{Y_{\text{control}}}\right)_i \mid \text{se}_i \sim \beta_0 + \beta_1 \log \text{rate} + \beta_2 \text{pH} + \beta_3 \text{MAT} + \beta_4 \text{MAP} + \beta_5 \log \text{grain} + \alpha_{\text{crop}} + \alpha_{\text{site}} + \varepsilon_i. \quad (6.2)$$

erw-yield-predict then maps the posterior-mean coefficients over the per-pixel covariates to a yield-ratio uplift raster, which erw-7 prefers over EcoCrop when present. At v1 the trial table has only ~ 14 rows, so the fit is prior-dominated: the framework is in place but the arm is **not run**, and EcoCrop is the operative response.

7. Cost, logistics, and energy

The cost stream builds the delivered-and-applied price of basalt pixel by pixel, and — because grinding and haulage burn diesel and electricity — the lifecycle emissions that must be netted off the gross CDR.

7.1 Feedstock supply and logistics (B4)

Basalt sources are the basic-volcanic, pyroclastic, and gabbro classes of the GLiM lithological map ([Hartmann & Moosdorf, 2012](#)); ultramafics are excluded for their nickel/chromium leaching risk. From every cropland pixel, `terra::costDist` computes the least-cost travel time to the nearest source over the Malaria Atlas motorised friction surface ([Weiss et al., 2020](#)). Travel time becomes distance and cost through simple, transparent constants ([Table 7.1](#)):

$$C_{\text{transport}} = t_{\text{travel}} \times 0.04 \text{ \$/min/t}, \quad d_{\text{km}} = t_{\text{travel}} \times 0.5, \quad (7.1)$$

$$C_{\text{bas/t}} = \underbrace{10}_{\text{quarry}} + \underbrace{C_{\text{grind}}}_{\text{energy}} + C_{\text{transport}} + \underbrace{8}_{\text{spreading}} \text{ (\$/t)}. \quad (7.2)$$

Haulage is capped at 1000 km, beyond which diesel emissions would erase the credit and cost makes basalt uncompetitive with lime. The delivered price surface is shown in [Figure 7.1](#).

Table 7.1: Logistics and cost constants.

Constant	Value	Note
Quarry-gate price	\$10/t	Ex-quarry basalt fines
Truck haulage	\$0.04/min/t	30-t truck at \$80/hr
Effective speed	0.5 km/min	≈30 km/h on-friction
Max haul	1000 km	Beyond this, unreachable
Spreading	\$8/t	On-farm application
Grinding (50 μm)	19.4 kWh/t	Equation 4.4
Truck emissions	0.12 kg CO ₂ /t/km	Diesel LCA
Spreading emissions	0.5 kg CO ₂ /t	≈0.1 L diesel/t

7.2 Energy: the largest lifecycle lever (B5)

Grinding cost and grinding emissions are both local: the same 19.4 kWh/t is priced at the country's industrial electricity tariff and emits at the country's grid carbon intensity. `erw-energy-country` rasterises a 43-country table. The grid carbon intensity spans roughly **0.03 kg CO₂/kWh** (hydro-rich Ethiopia, DR Congo) to **0.95 kg CO₂/kWh** (coal-based South Africa) — a $\sim 30\times$ spread that makes electricity, not any weathering parameter, the single largest control on *lifecycle* emissions.

7.3 Lifecycle emissions

Per tonne of basalt, lifecycle emissions sum grinding, transport, and spreading:

$$L = \underbrace{(\text{kWh/t}) \cdot \text{CI}}_{\text{grinding}} + \underbrace{d_{\text{km}} \cdot 0.12}_{\text{transport}} + \underbrace{0.5}_{\text{spreading}} \quad (\text{kg CO}_2/\text{t}), \quad (7.3)$$

with CI the grid carbon intensity. This L is what the profitability engine deducts from gross CDR to get the net removal that earns credits.

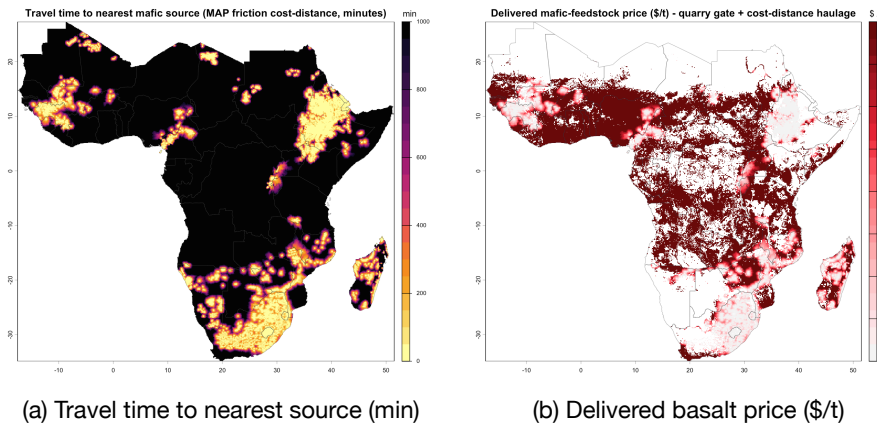


Figure 7.1: Logistics surfaces. Delivered price is lowest near basalt outcrops and rises with haul distance; it is the gate that decides which otherwise profitable pixels actually clear.

8. Profitability

The profitability engine (erw-7) assembles the three streams into [Equation 2.1](#) for every crop, allocation, and regime — 276 output rasters in all.

8.1 The gross-margin identity, concretely

$$\text{Rev}_{\text{CDR}} = \text{CDR}_{\text{net}} \times (p_{\text{CO}_2} - \text{MRV}), \quad (8.1)$$

$$\text{CDR}_{\text{net}} = \max(0, \text{CDR}_{\text{gross}} - R L / 1000), \quad (8.2)$$

$$\text{Rev}_{\text{agro}} = \Delta Y \times p_c, \quad C_{\text{basalt}} = R \times C_{\text{bas/t}}, \quad (8.3)$$

$$\text{GM}_{\text{combined}} = \text{Rev}_{\text{agro}} + \text{Rev}_{\text{CDR}} - C_{\text{basalt}}. \quad (8.4)$$

The engine also reports agro-only and CDR-only margins and a return-on-investment $\text{ROI} = (\text{Rev}_{\text{agro}} + \text{Rev}_{\text{CDR}}) / C_{\text{basalt}}$. Default economic constants are in [Table 8.1](#).

Table 8.1: Default economic and carbon-market constants (erw-7).

Constant	Value	Source
Carbon price p_{CO_2}	\$150/tCO ₂	VCM mid 2024–25 (Levy et al., 2024)
MRV cost	\$20/tCO ₂	Levy et al. (2024)
Discount rate	10%	Conventional
Project horizon	5 yr	Accreditation window
CDR phasing	0.30/0.25/0.20/0.15/0.10	Kanzaki et al. (2023)
Grid CI fallback	0.60 kg/kWh	Mirzabaev et al. (2019)

8.2 Three temporal accounting regimes

The same pixel can be scored three ways, reflecting genuinely different questions about *when* the returns land ([Table 8.2](#)).

Table 8.2: The three return regimes.

Regime	Definition
year1	One application; agronomic return accumulated <i>undiscounted</i> over the 5-year residency to match the horizon-cumulative CDR: $GM = Rev_{agro} \times 5 + Rev_{CDR} - C_{basalt}$.
npv	Agronomic return discounted at 10% over the reapplication interval (via <code>limer::NPV_lime</code>); CDR revenue multiplied by the CDR-NPV factor ≈ 0.79 .
equilibrium	Steady state: the annual <i>maintenance</i> rate and its CDR, no horizon multiplier and no discount factor.

The CDR-NPV factor discounts the phased removal relative to treating it as upfront:

$$\text{CDR-NPV factor} = \frac{\sum_t f_t / (1 + r)^t}{\sum_t f_t} \approx 0.79, \quad (8.5)$$

with $r = 10\%$ and the phasing $f = \{0.30, 0.25, 0.20, 0.15, 0.10\}$.

8.3 Allocation rules and breakeven carbon price

Four allocation rules are scored: `targeted` (the LiTAS dose, only on pixels acidic beyond each crop's tolerance) and `uniform_10/20/50`. The `targeted` rule is the parsimonious one — it applies basalt only where the soil actually needs it — while the uniform rules mirror the continental-study convention and expose the over-application penalty ([chapter 9](#)).

Rearranging [Equation 8.4](#) for the carbon price that drives the crop-weighted margin to zero gives the *breakeven carbon price*, a map of how high the carbon market must be for ERW to clear at each location:

$$p_{CO_2}^{\text{breakeven}} = \frac{C_{\text{basalt}} - Rev_{\text{agro}}}{CDR_{\text{net}}} + MRV. \quad (8.6)$$

[Figure 8.1](#) shows the year-1 `targeted` gross margin and its breakeven surface.

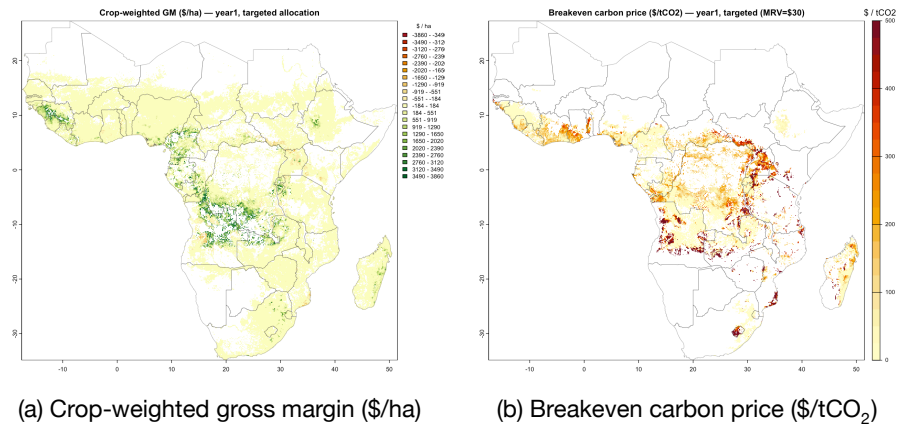


Figure 8.1: Profitability outcome maps, year-1 targeted allocation.

Worked intuition (pre-refresh defaults, for teaching). Pixel A near Kisumu, Kenya (pH 5.2, warm and wet, 150 km haul) reaches $CDR_{px} \approx 172 \text{ kg/t}$ and $\sim 4.6 \text{ tCO}_2/\text{ha}$ net; Pixel B near Kaya, Burkina Faso (pH 6.0, semi-arid, 500 km haul) reaches only $\sim 58 \text{ kg/t}$ — and there the lifecycle emissions exceed the gross capture, so net CDR is zero. The lesson: basalt cost dominates, climate is the second-largest lever, and *which pixel* you pick matters more than the average price.

9. Supply curves and sensitivity

9.1 Supply ranking (B8)

When basalt supply is the scarce input, the right way to deploy a limited annual tonnage is to rank pixels by *gross margin per tonne of basalt* — the shadow value of the supply constraint — and fill from the top until the cap binds. *erw-supply* does exactly this, producing a supply curve of marginal \$/t basalt against annual capacity and the cumulative carbon it buys.

Ranking and cap matter as much as parameters. Because the profitable pixels are a minority of a heavy-tailed distribution, a well-chosen top decile is strongly positive even where the SSA-wide *mean* is negative. The deployment question is therefore “which pixels, up to what cap?” at least as much as “what carbon price?”.

9.2 Kinetic saturation makes CDR concave in dose

Applying more basalt does not buy proportionally more durable carbon: the soil’s acidity sink is filled first ([Equation 5.5](#)), and the reactive fraction saturates. Durable net-export CDR is therefore concave — even non-monotone — in the application rate ([Figure 9.1](#)), which is precisely why the uniform-50 rule destroys value.

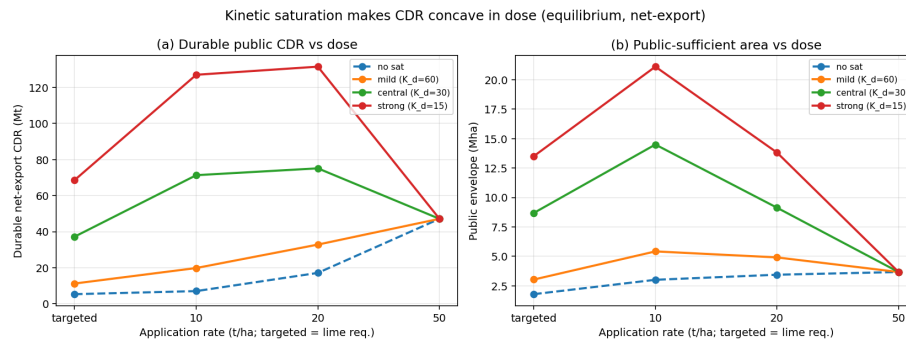


Figure 9.1: Kinetic saturation: durable public CDR and the public-sufficient area are concave in dose. Past a point, extra basalt adds cost and lifecycle emissions faster than durable removal.

9.3 Sensitivity sweeps (B9)

erw-8 runs five one- or two-dimensional sweeps over the whole engine, across all three regimes and 23 crops ([Table 9.1](#)).

Table 9.1: Sensitivity sweeps.

Sweep	Grid
Prices	crop $\times$ basalt price multipliers $\{0.5 \dots 2.0\}$, carbon $\{0, 50, 100, 150, 250\}$
Yields	yield-gap factor $\{1.0, 1.25, \dots, 2.5\}$
MRV	$\{0, 15, 30, 50, 80\}$ $\$/\text{tCO}_2$
Grain	$\{10, 30, 50, 100, 200, 500\}$ μm (reactivity vs. grinding)
Allocation	targeted, uniform 10/20/50

The over-application penalty is visible in [Figure 9.2](#): private return falls steadily from the targeted (lime-requirement) dose through the uniform ladder, and the over-liming sensitivity band widens as the dose grows.

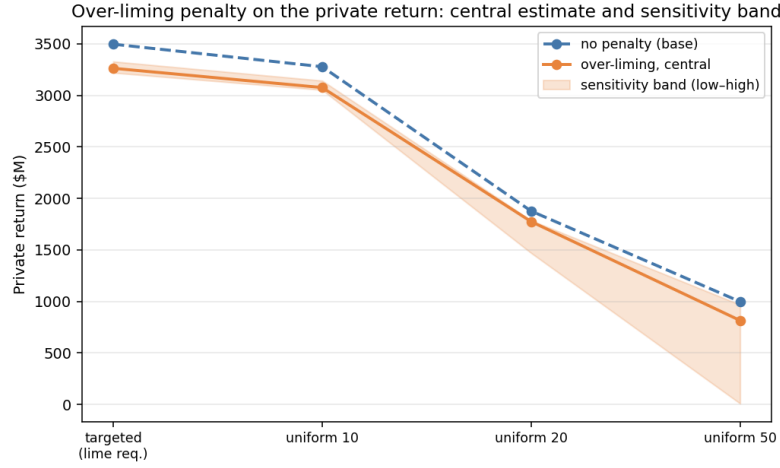


Figure 9.2: Over-application penalty on the private return across allocation rules. The targeted (lime-requirement) dose dominates; uniform-50 gives up most of the value.

9.4 Private, public, and combined returns

A dedicated analysis (erw-11) asks a sharper question than “is the pixel profitable?”: *which* of the two returns carries it. Because erw-7 already separates the agronomic and CDR margins — each measured against the *full* delivered basalt cost — every pixel can be classified by three independent sufficiency tests:

$$\text{private-sufficient} \iff \text{GM}_{\text{agro-only}} = R_{\text{agro}} - C_{\text{basalt}} > 0, \quad (9.1)$$

$$\text{public-sufficient} \iff \text{GM}_{\text{CDR-only}} = R_{\text{CDR}} - C_{\text{basalt}} > 0, \quad (9.2)$$

$$\text{combined-sufficient} \iff \text{GM}_{\text{combined}} = R_{\text{agro}} + R_{\text{CDR}} - C_{\text{basalt}} > 0. \quad (9.3)$$

Private-sufficient land pays for itself on yield alone — a grower would invest with or without carbon finance; public-sufficient land pays on carbon revenue alone — a buyer would fund it with or without yield gains. Their *intersection* is doubly justified and, crucially, is exactly where carbon credits are *least additional*, because the agronomic case already stands on its own. This yields a five-way per-pixel typology: **both**, **private-only**, **public-only**, **combined-only** (neither stream alone suffices but stacked they do), and **neither**.

9.4.1 Robustness and the core targeting area

Because the intersection is parameter-sensitive, erw-11 recomputes it analytically over a 240-point grid — delivered cost $\times \{0.5-2.0\}$, carbon price

$\{\$50\text{--}250\}$, $\text{yield} \times \{0.5\text{--}2.0\}$, $\text{CDR rate} \times \{0.5\text{--}1.5\}$ — and scores each pixel by the fraction of that grid over which it remains in the intersection. The **core** targeting area is the land that survives $\geq 50\%$ of the grid; **candidate** land survives $\geq 33\%$. A separate plausible-central-band cross-check gives an independent robust-core estimate.

9.4.2 What binds the public envelope

Public-sufficiency ([Equation 9.2](#)) reduces to a marginal-abatement-cost test in which the per-hectare dose cancels:

$$\text{MAC} = \frac{C_{\text{bas/t}}}{\text{CDR}_{\text{per-t}}} \quad (\$/\text{tCO}_2), \quad \text{public-sufficient} \iff \text{MAC} < p_{\text{CO}_2} - \text{MRV}. \quad (9.4)$$

Because realised durable CDR is only $\sim 0.1\text{--}0.25 \text{ tCO}_2$ per tonne of basalt, the MAC sits *above* typical voluntary-market carbon prices on most cropland. The operational conclusion is that **the realised CDR rate, not the credit price, is the binding constraint** — levers that raise the removal per tonne (finer grain, warmer/wetter siting, better feedstock) move the frontier more than a higher carbon price does (quantified in [section 10.4](#)).

10. Results

Headline numbers below are from the run dated 2026-05-26 (carbon \$150, MRV \$20, 50 μm , spatial logistics, country energy, per-crop EcoCrop pH cliffs).

10.1 Continent-wide totals

Table 10.1: SSA-wide totals by regime and allocation. GM = gross margin; CDR = durable removal; profitable area where $\text{GM} > 0$.

Regime	Allocation	GM (\$M)	CDR rev (\$M)	CDR (Mt)	Prof. (Mha)	CDR on prof. (Mt)
year-1	targeted	+18,440	10,566	81.3	137.9	58.4
NPV	targeted	−1,107	8,391	81.3	132.4	37.4
equilibrium	targeted	+1,965	3,442	26.5	134.9	13.6
year-1	uniform 10	+19,950	13,742	105.7	127.1	81.6
NPV	uniform 10	−6,106	10,914	105.7	119.2	68.3
year-1	uniform 20	+9,712	27,485	211.4	124.1	153.7
NPV	uniform 20	−17,701	21,827	211.4	117.4	130.4
year-1	uniform 50	−21,002	68,712	528.6	120.6	355.1
NPV	uniform 50	−52,486	54,567	528.6	116.1	314.6
equilibrium	uniform 50	−45,153	68,712	528.6	117.1	325.4

Reading [Table 10.1](#): year-1 targeted leads at **+\$18.4B** on 137.9 Mha; uniform-10 is marginally higher in absolute terms but at far greater basalt spend; NPV-targeted is only mildly negative (−\$1.1 B); and uniform-50 is negative in every regime — the over-application penalty in numbers. The number of crops with a positive pixel-mean margin rises from 1 (pre-refresh) to 12 (year-1 targeted) and 19 (equilibrium targeted). The committed *area* totals here are flagged stale relative to the current rasters (the current-data footprint is single-digit Mha; see the caveat in [section 10.4](#)), so read this table for the relative ranking of regimes and allocations rather than the absolute hectares.

10.2 Which crops clear

Table 10.2: Top crops by LiTAS-targeted pixel-mean gross margin (\$/ha).

Crop	Year-1 (5-yr cum.)	NPV	Equilibrium
Potato	10,560	3,017	1,984
Tobacco	4,884	807	798
Sweet potato	2,867	358	450
Lentil	2,628	885	491
Groundnut	2,216	377	401
Cassava	2,018	—27	170
Common bean	1,990	286	359
Chickpea	1,719	368	299
Wheat	1,662	74	225

High-value roots, tubers, and legumes on acidic soils clear most easily; acid-tolerant cereals (sorghum, millet) and the tree commodities (coffee, cocoa) stay negative because their EcoCrop response is already saturated, leaving carbon revenue to shoulder the entire basalt cost.

10.3 Supply curves

Table 10.3: Marginal \$/t basalt at selected annual caps, and the peak total gross margin.

Regime	10 Mt	50 Mt	100 Mt	250 Mt	Peak total GM
year-1	+186	+117	+79	+21	\$19,648M @ 250 Mt
NPV @10%	+86	+46	+16	—16	\$5,018M @ 100 Mt
equilibrium	+91	+20	—14	—60	\$3,145M @ 100 Mt

The marginal pixel turns unprofitable at roughly 250 Mt/yr (year-1), 100 Mt/yr (NPV), and 50–100 Mt/yr (equilibrium) — an economically meaningful deployment frontier well below the physically available tonnage.

10.4 Private, public, and combined envelopes

Applying the sufficiency tests of [section 9.4 \(Equation 9.1–Equation 9.3\)](#) to the baseline (year-1 targeted, \$150/tCO₂, MRV \$20, area-weighted across the 23 crops sharing each pixel) gives the three envelopes in [Table 10.4](#).

Table 10.4: Private, public, and combined return envelopes: area-weighted footprint, farms reached, gross value of crop production on the affected land, and cumulative net CDR (erw-11, year-1 baseline).

Allocation	Envelope	Area (Mha)	Farms (M)	Value (\$M)	Net CDR (Mt)
Targeted	Private	8.22	6.5	8,590	47.9
Targeted	Public	2.23	1.6	1,680	25.0
Targeted	Intersection	1.85	1.3	1,486	22.2
Targeted	Combined > 0	11.95	9.1	10,761	65.3
Uniform 20	Private	7.98	6.4	8,966	30.1
Uniform 20	Public	3.51	2.6	2,746	17.8
Uniform 20	Intersection	1.97	1.3	1,669	10.7
Uniform 20	Combined > 0	13.12	10.2	13,955	51.5

Decomposing the targeted case into the five-way typology (harvested area): **private-only 6.37 Mha** (yield gains alone justify investment — carbon credits here are non-additional), **public-only 0.38 Mha**, **both (intersection) 1.85 Mha**, and **combined-only 3.35 Mha** (neither stream alone suffices, but stacked they do — the band where blended carbon-plus-agronomic finance is genuinely pivotal). The headline tension is that the private case is far larger than the public case (8.2 vs. 2.2 Mha): most agronomically-attractive land would *not* depend on carbon credits, while the land where carbon finance is decisive is precisely where the private case falls short. Allocation mostly trades CDR depth for footprint — uniform-20 widens the combined footprint (13.1 vs. 12.0 Mha) but roughly halves intersection CDR (10.7 vs. 22.2 Mt) by spreading a fixed dose onto near-neutral soils with a weak agronomic response.

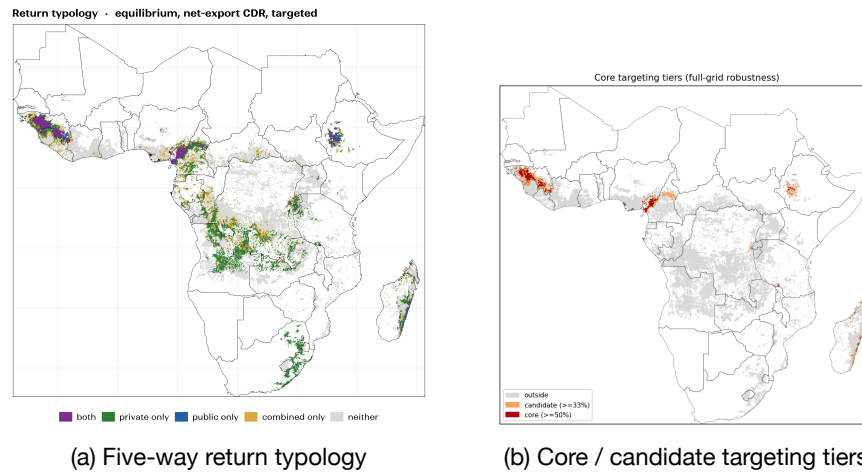


Figure 10.1: Where each return carries the investment (left) and the robustness-based core targeting area (right). Maps use the conservative equilibrium, net-export accounting; the core concentrates in Cameroon, Guinea, the Ethiopian highlands, and coastal Madagascar.

10.4.1 The robust core

No pixel stays in the intersection across more than 57% of the 240-point robustness grid — the doubly-justified case is inherently parameter-sensitive. On the robustness gradient (Table 10.5), a plausible-central-band cross-check independently gives ~ 0.81 Mha, so the two methods converge on $\sim 0.7\text{--}0.8$ Mha of genuinely robust core land. It concentrates where acidic soils, warm-wet climate, basalt proximity, and decent crop prices coincide: Cameroon (0.42 Mha core), Guinea (0.25), the Ethiopian highlands (0.05), and Tanzania (0.02).

Table 10.5: Robustness-based targeting tiers (targeted allocation).

Tier	Definition	Area (Mha)	Farms (M)	Value (\$M)	CDR (Mt)
Core	intersection under $\geq 50\%$ of grid	0.74	0.45	663	13.1
Candidate	intersection under $\geq 33\%$ of grid	1.73	1.16	1,409	21.5

10.4.2 What binds the public envelope

The public envelope is small because durable removal is expensive per tonne, not because the credit price is low. The area-weighted marginal-abatement cost (Equation 9.4) across treated cropland is **p25 \$247, median \$305, p75 \$476/tCO₂** (Figure 10.2) — set against a net credit price of only \$130 (\$150 less \$20 MRV). Only the cheapest $\sim 10\text{--}15\%$ of land clears, which is why the public envelope is just 2.2 Mha, and the price ladder is steeply convex (Table 10.6).

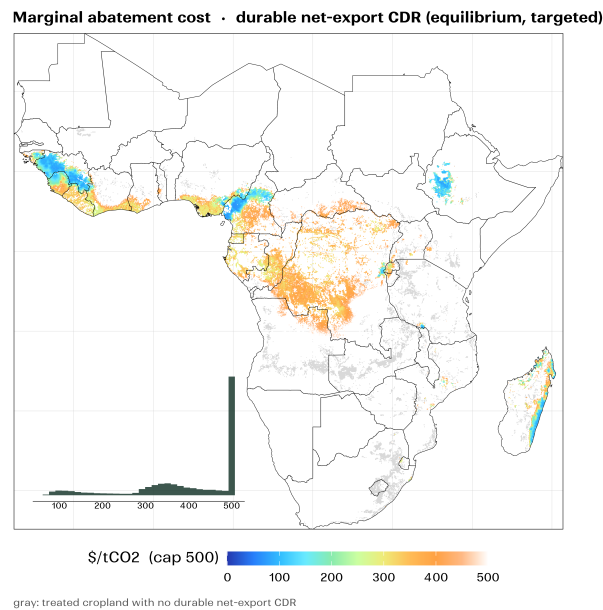


Figure 10.2: Marginal abatement cost of durable net-export CDR. The area-weighted distribution (inset histogram) sits well above the \$130/tCO₂ net credit price, so only a small, cheap fraction of cropland is public-sufficient.

Table 10.6: Public-sufficient area and cumulative CDR as the VCM price rises.

VCM price (\$/tCO ₂)	Public area (Mha)	Cumulative CDR (Mt)
150 (current)	2.23	25
250	3.57	31
300	7.52	51
500	13.15	73
1000	16.6	81

A lever comparison settles which knob matters: +33% carbon price widens the public envelope by +26%, –33% delivered cost by +33%, +33% CDR rate by +23% — but **doubling the realised CDR rate widens it by +181%** (2.23 → 6.27 Mha), beyond any plausible price or cost move. And a one-at-a-time tornado on the intersection area confirms **delivered basalt cost is the single dominant lever** (swing 5.62 Mha over $\times 0.5$ –2.0), larger than carbon price (2.80), CDR rate (2.38), and yield (0.89) — so targeting is first a logistics problem. Reassuringly, the yield benefit is the *least* sensitive input, which matters because it is also the most uncertain (the Bayesian arm is prior-dominated).

Caveat on absolute area. The committed output-ssa-summary.csv figures reproduced in [Table 10.1](#) (e.g. 137.9 Mha “profitable”) are flagged *stale* relative to the current economics_erw rasters: recomputing erw-10’s own definition on today’s data gives ~ 9.9 Mha combined-positive, consistent with the single-digit-Mha envelope areas above. Read [Table 10.1](#) for the *relative* ranking of regimes and allocations, and this section for the current-data footprint. Year-1 is also the most optimistic regime; NPV and equilibrium shrink every envelope.

10.5 What the results say

1. **Targeted beats uniform.** Dosing to lime requirement dominates flat application in every regime; uniform-50 destroys value.
2. **The CDR rate binds, not the carbon price** ([Equation 9.4](#)): realised removal per tonne is small, so most cropland needs either a co-benefit or a cheaper/finer feedstock to clear on carbon alone.
3. **Private and public returns are complementary** ([Figure 5.3](#)): the same acidity gradient that maximises yield relief minimises exported carbon, and vice versa.
4. **The private case is larger than the public case** ([section 10.4](#)): 8.2 vs. 2.2 Mha targeted. Carbon finance is decisive mainly on the 3.35 Mha combined-only band, and *least additional* on the doubly-justified intersection — with a robust core of only ~ 0.7 – 0.8 Mha.
5. **Ranking and cap dominate the average:** a positive top decile exists even where the mean is negative.

11. Assumptions, limitations, and roadmap

11.1 Where the model uses defensible proxies (v1 flags)

The CDR stream in particular is a first-order parameterisation. Each proxy below is a deliberate, sourced simplification, not an oversight:

1. **Linear climate multipliers** (Equation 5.1) stand in for full reactive-transport kinetics; to be replaced by a Kanzaki-class form when calibration data allow (Kanzaki et al., 2023).
2. **Triangular pH factor** peaked at pH 5 (Equation 5.2) is a defensible midpoint, not a calibrated curve.
3. **Reactive fraction** $\phi_0 = 0.20$ is anchored to a single temperate mesocosm (Lewis et al., 2021).
4. **Grain-size exponent** $\beta = 0.5$ is the midpoint of an empirical 0.35–0.5 range, chosen to avoid a free parameter.
5. **Pedogenic loss** (Table 5.1) uses aridity bands, with MAP as a fallback proxy for MAP/PET that over-deducts in cool highlands.
6. **EcoCrop as the operative yield response** captures tolerance ranges, not basalt-specific uplift; the trial-anchored Bayesian arm is not yet run.

11.2 What the model does not represent

N₂O and soil-compaction emissions from heavy application; Article 6 / credit eligibility and corresponding adjustments; land tenure and who finances the ~\$600/ha upfront cost; industrial alkaline by-products (steel slag, cement kiln dust) that could supply feedstock in basalt-poor regions; ultramafic sources (excluded on safety grounds); a spatially varying MRV cost (a single flat \$20/tCO₂ is used); and any aggregator-level, as opposed to farmer-level, economics.

11.3 Roadmap: from v0 to v2

This report documents a **v0 screening model**: transparent, fast, and honest about its proxies. It is meant to rank and compare, not to certify.

- v0 (this report)** Interpretable multipliers; gross *and* net bounds carried side by side; every sensitive parameter swept. Good for “where” and “how sensitive”.
- v1** Replace the reactive-fraction/climate block with an analytical shrinking-core, particle-size-distribution dissolution model; run the Bayesian yield arm on an expanded trial set.
- v2** Emulate a mechanistic 1-D reactive-transport model (SCEPTER-class) over a designed experiment, and evaluate the fast surrogate per pixel — the path to credit-grade absolute numbers.

Because the v0 already carries gross and net bounds rather than a single point estimate, and because its errors are largely monotone across the map, its *ranking* of locations is expected to be robust even as absolute tonnages are refined — which is exactly what a screening model needs to be right about.

12. Reproducibility and provenance

12.1 Run order

```
erw-0    install packages
erw-1    soil layers + SSA/cropland grid
erw-2    SPAM crop layers
erw-3    feedstock chemistry + basalt rate
erw-4    crop EcoCrop parameter tables
erw-5    EcoCrop relative-yield rasters
erw-9    CDR surface (climate, pedogenic, net-export)
erw-basalt-access  logistics / delivered price
erw-energy-country  electricity + grid CI
[erw-yield-fit, erw-yield-predict] optional Bayesian arm
erw-7    profitability (276 rasters)
erw-8    sensitivity sweeps
erw-supply  supply curves + deployment masks
erw-10, erw-11, erw-plot-maps  maps and tables
```

12.2 Environment and outputs

R 4.5+ with terra, geodata, sf, here, brms, malariaAtlas, and the GAIA limer and Recocrop packages. The profitability engine writes `eco-nomics_erw/{allocation}/{crop}_{regime}.tif`, each carrying the full decomposition (area, yield, loss, agronomic return, basalt rate and cost, gross and net CDR, CDR revenue, and the combined / agro-only / CDR-only margins). The constants catalogue with citations is `erw-parameters.md`; the CDR derivation is `erw-cdr-calculations.pdf`.

References

- Aramburu Merlos, F., Silva, J. V., Baudron, F., & Hijmans, R. J. (2023). Estimating lime requirements for tropical soils: Model comparison and development. *Geoderma*, 432, 116421. (<https://doi.org/10.1016/j.geoderma.2023.116421>)
- Baek, S. H., Kanzaki, Y., Lora, J. M., Planavsky, N., Reinhard, C. T., & Zhang, S. (2023). Impact of climate on the global capacity for enhanced rock weathering on croplands. *Earth's Future*, 11, e2023EF003698. (<https://doi.org/10.1029/2023EF003698>)
- Beerling, D. J., Kantzas, E. P., Lomas, M. R., Wade, P., Eufrazio, R. M., Renforth, P., ... others (2020). Potential for large-scale CO₂ removal via enhanced rock weathering with croplands. *Nature*, 583, 242–248. (<https://doi.org/10.1038/s41586-020-2448-9>)
- Brantley, S. L., Kubicki, J. D., & White, A. F. (2008). Kinetics of water-rock interaction. In S. L. Brantley, J. D. Kubicki, & A. F. White (Eds.), *Kinetics of water-rock interaction*. New York: Springer. (<https://doi.org/10.1007/978-0-387-73563-4>)
- Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, 80(1), 1–28. (<https://doi.org/10.18637/jss.v080.i01>)
- Dietzen, C., & Rosing, M. T. (2023). Quantification of CO₂ uptake by enhanced weathering of silicate minerals applied to acidic soils. *International Journal of Greenhouse Gas Control*, 125, 103872. (<https://doi.org/10.1016/j.ijggc.2023.103872>)
- Dupla, X., Bertagni, M. B., & Grand, S. (2025). Three years of field trials indicate a sustained enhanced rock weathering signal with limited CO₂ removal. *Environmental Science & Technology*, 59(48), 25751–25764. (<https://doi.org/10.1021/acs.est.5c09820>)
- FAO. (2023). *FAOSTAT producer prices*. Food and Agriculture Organization of the United Nations. (<https://www.fao.org/faostat/>)
- Fick, S. E., & Hijmans, R. J. (2017). WorldClim 2: new 1-km spatial resolution climate surfaces for global land areas. *International Journal of Climatology*, 37(12), 4302–4315. (<https://doi.org/10.1002/joc.5086>)
- Gíslason, S. R., & Oelkers, E. H. (2003). Mechanism, rates, and consequences of basaltic glass dissolution: II. an experimental study of the dissolution rates of basaltic glass as a function of pH and temperature. *Geochimica et Cosmochimica Acta*, 67(20), 3817–3832. ([https://doi.org/10.1016/S0016-7037\(03\)00176-5](https://doi.org/10.1016/S0016-7037(03)00176-5))
- Gudbrandsson, S., Wolff-Boenisch, D., Gíslason, S. R., & Oelkers, E. H. (2011).

- An experimental study of crystalline basalt dissolution from $2 \leq \text{pH} \leq 11$ and temperatures from 5 to 75 °C. *Geochimica et Cosmochimica Acta*, 75(19), 5496–5509. (<https://doi.org/10.1016/j.gca.2011.06.035>)
- Hamilton, S. K., Kurzman, A. L., Arango, C., Jin, L., & Robertson, G. P. (2007). Evidence for carbon sequestration by agricultural liming. *Global Biogeochemical Cycles*, 21, GB2021. (<https://doi.org/10.1029/2006GB002738>)
- Harrington, K. J., Hilton, R. G., & Henderson, G. M. (2023). Implications of the riverine response to enhanced weathering for CO₂ removal in the UK. *Applied Geochemistry*, 152, 105643. (<https://doi.org/10.1016/j.apgeochem.2023.105643>)
- Hartmann, J., & Moosdorf, N. (2012). The new global lithological map database GLiM: A representation of rock properties at the Earth surface. *Geochemistry, Geophysics, Geosystems*, 13(12), Q12004. (<https://doi.org/10.1029/2012GC004370>)
- Hartmann, J., West, A. J., Renforth, P., Köhler, P., De La Rocha, C. L., Wolf-Gladrow, D. A., ... Scheffran, J. (2013). Enhanced chemical weathering as a geoengineering strategy to reduce atmospheric carbon dioxide, supply nutrients, and mitigate ocean acidification. *Reviews of Geophysics*, 51(2), 113–149. (<https://doi.org/10.1002/rog.20004>)
- Hijmans, R. J. (2023). *Recocrop: Environmental suitability for plants using the EcoCrop model*. R package. (<https://CRAN.R-project.org/package=Recocrop>)
- Kantola, I. B., Blanc-Betes, E., Masters, M. D., Chang, E., Marklein, A., Moore, C. E., ... DeLucia, E. H. (2023). Improved net carbon budgets in the U.S. midwest through direct measured impacts of enhanced weathering. *Global Change Biology*, 29(24), 7012–7028. (<https://doi.org/10.1111/gcb.16903>)
- Kanzaki, Y., Planavsky, N. J., & Reinhard, C. T. (2023). New estimates of the storage permanence and ocean co-benefits of enhanced rock weathering. *PNAS Nexus*, 2(4), pgad059. (<https://doi.org/10.1093/pnasnexus/pgad059>)
- Larkin, C. S., Andrews, M. G., Pearce, C. R., Yeong, K. L., Beerling, D. J., Bellamy, J., ... James, R. H. (2022). Quantification of CO₂ removal in a large-scale enhanced weathering field trial on an oil palm plantation in sabah, malaysia. *Frontiers in Climate*, 4, 959229. (<https://doi.org/10.3389/fclim.2022.959229>)
- Levy, C. R., Almaraz, M., Beerling, D. J., Raymond, P., Reinhard, C. T., Suhrhoff, T. J., & Taylor, L. (2024). Enhanced rock weathering for carbon removal—monitoring and mitigating potential environmental impacts on agricultural land. *Environmental Science & Technology*, 58(39), 17215–17226. (<https://doi.org/10.1021/acs.est.4c02368>)
- Lewis, A. L., Sarkar, B., Wade, P., Kemp, S. J., Hodson, M. E., Taylor, L. L., ... others (2021). Effects of mineralogy, chemistry and physical properties of basalts on carbon capture potential and plant-nutrient element

- release via enhanced weathering. *Applied Geochemistry*, 132, 105023. (<https://doi.org/10.1016/j.apgeochem.2021.105023>)
- Mirzabaei, A., Wu, J., Evans, J., García-Oliva, F., Hussein, I. A. G., Iqbal, M. H., ... Weltz, M. (2019). Desertification. In P. R. Shukla, J. Skea, et al. (Eds.), *Climate change and land: An IPCC special report (SRCCL)* (chap. 3). Intergovernmental Panel on Climate Change. (<https://www.ipcc.ch/srccl/chapter/chapter-3/>)
- Palandri, J. L., & Kharaka, Y. K. (2004). *A compilation of rate parameters of water-mineral interaction kinetics for application to geochemical modeling* (Open-File Report No. 2004-1068). U.S. Geological Survey. (<https://pubs.usgs.gov/publication/ofr20041068>)
- Poggio, L., de Sousa, L. M., Batjes, N. H., Heuvelink, G. B. M., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *SOIL*, 7(1), 217–240. (<https://doi.org/10.5194/soil-7-217-2021>)
- Prohaska, T., Irrgeher, J., Benefield, J., Bohlke, J. K., Chesson, L. A., Coplen, T. B., ... others (2022). Standard atomic weights of the elements 2021 (IUPAC technical report). *Pure and Applied Chemistry*, 94(5), 573–600. (<https://doi.org/10.1515/pac-2019-0603>)
- Reershemius, T., Kelland, M. E., Jordan, J. S., Davis, I. T., D’Ascanio, R., Kalderon-Asael, B., ... Planavsky, N. J. (2023). Initial validation of a soil-based mass-balance approach for empirical monitoring of enhanced rock weathering rates. *Environmental Science & Technology*, 57(48), 19497–19507. (<https://doi.org/10.1021/acs.est.3c03609>)
- Reershemius, T., & Suhrhoff, T. J. (2024). On error, uncertainty, and assumptions in calculating carbon dioxide removal rates by enhanced rock weathering. *Global Change Biology*, 30(1), e17025. (<https://doi.org/10.1111/gcb.17025>)
- Renforth, P. (2012). The potential of enhanced weathering in the UK. *International Journal of Greenhouse Gas Control*, 10, 229–243. (<https://doi.org/10.1016/j.ijggc.2012.06.011>)
- Renforth, P. (2019). The negative emission potential of alkaline materials. *Nature Communications*, 10, 1401. (<https://doi.org/10.1038/s41467-019-09475-5>)
- Strefler, J., Amann, T., Bauer, N., Kriegler, E., & Hartmann, J. (2018). Potential and costs of carbon dioxide removal by enhanced weathering of rocks. *Environmental Research Letters*, 13(3), 034010. (<https://doi.org/10.1088/1748-9326/aaa9c4>)
- Suhrhoff, T. J., Reershemius, T., Wang, J., Jordan, J. S., Reinhard, C. T., & Planavsky, N. J. (2024). Measuring enhanced weathering: inorganic carbon-based approaches may be required to complement cation-based approaches. *Frontiers in Climate*, 6, 1352825. (<https://doi.org/10.3389/fclim.2024.1352825>)
- United Nations Environment Programme. (1992). *World atlas of desertification*. London: Edward Arnold.
- Weiss, D. J., Nelson, A., Vargas-Ruiz, C. A., Gligorić, K., Bavadekar, S.,

- Gabrilovich, E., ... others (2020). Global maps of travel time to health-care facilities. *Nature Medicine*, 26, 1835–1838. (<https://doi.org/10.1038/s41591-020-1059-1>)
- West, T. O., & McBride, A. C. (2005). The contribution of agricultural lime to CO₂ emissions in the united states: dissolution, transport, and net emissions. *Agriculture, Ecosystems & Environment*, 108(2), 145–154. (<https://doi.org/10.1016/j.agee.2005.01.002>)
- White, A. F., & Blum, A. E. (1995). Effects of climate on chemical weathering in watersheds. *Geochimica et Cosmochimica Acta*, 59(9), 1729–1747. ([https://doi.org/10.1016/0016-7037\(95\)00078-E](https://doi.org/10.1016/0016-7037(95)00078-E))
- White, A. F., & Brantley, S. L. (2003). The effect of time on the weathering of silicate minerals: why do weathering rates differ in the laboratory and field? *Chemical Geology*, 202(3–4), 479–506. (<https://doi.org/10.1016/j.chemgeo.2003.03.001>)
- Yu, Q., You, L., Wood-Sichra, U., Ru, Y., Joglekar, A. K. B., Fritz, S., ... Yang, P. (2020). A cultivated planet in 2010—Part 2: The global gridded agricultural-production maps. *Earth System Science Data*, 12(4), 3545–3572. (<https://doi.org/10.5194/essd-12-3545-2020>)
- Zhang, S., Planavsky, N. J., Katchinoff, J., Raymond, P. A., Kanzaki, Y., Reershemius, T., & Reinhard, C. T. (2022). River chemistry constraints on the carbon capture potential of surficial enhanced rock weathering. *Limnology & Oceanography*, 67(S1), S148–S157. (<https://doi.org/10.1002/lno.12244>)
- Zomer, R. J., Xu, J., & Trabucco, A. (2022). Version 3 of the global aridity index and potential evapotranspiration database. *Scientific Data*, 9, 409. (<https://doi.org/10.1038/s41597-022-01493-1>)